
Minsu Kim

Korea Advanced Institute of Science and Technology (KAIST),
Electrical Engineering, Ph.D Student.

Phone.: +8210-6406-2448,
Email : minshu.kim@kaist.ac.kr,
Address: Yuseong-gu, Daejeon, Republic of Korea.

Education

- **Ph.D. (2025.02 -)**, Korea Advanced Institute of Science & Technology (KAIST), Electrical Engineering, Republic of Korea.
 - Thesis: -
 - Keyword: Efficient/On-Device AI.
- **M. S. (2021.08 - 2023.08)**, Daegu Gyeongbuk Institute of Science & Technology (DGIST), Electrical Engineering and Computer Science, Republic of Korea.
 - Thesis: Enhanced Image Stitching using Recurrent Elastic Warps and High-Definition Neural Imaging
 - Keyword: Representation Learning for Panorama Stitching.
- **B. S. (2015.02 - 2021.08)**, Pusan National University (PNU), Electrical Engineering, Republic of Korea.
 - Graduation Project: Robot Vision and Mobile Manipulation for Autonomous Cleaning Robot
 - Keyword: Robotics (Vision & Control).

Work Experience

- **Vision & AI Specialist (2023.12 - 2025.01)**, KIST, Republic of Korea.
 - Topic: Robot system integration, computer vision, and image processing for autonomous robots.
 - Keyword: ROS2, C++, Python, Cuda Programming, Embedded S/W, On-device AI
 - Achievement: Development of an intelligent quadruped robot.
- **Research Intern (2023.06 - 2023.12)**, NAVER LABS, Republic of Korea.
 - Topic: Bird's Eye View Perception.
 - Keyword: 3D Detection, Semantic Segmentation.
 - Achievement: Publication of 2 Papers to flagship venues (ICRA, RA-L).
- **Research Intern (2019.12 - 2021.08)**, Pusan National University, Autonomous Robot & Artificial Intelligence (ARAI) Lab, Republic of Korea.
 - Topic: Robot Vision & Control.
 - Keyword: ROS, C++, Python, Embedded S/W.
 - Achievement: Started with zero-base, became a winner of a national league in one year.

Research Interest

- **Image Processing**
 - Super-resolution, Warping, Image Blending, Panorama stitching.
- **Machine Learning for Robotics**
 - Semantic Segmentation, Detection, Classification, Bird's Eye View Perception.
- **Efficient Deep Learning**
 - Model Quantization, Model Pruning, Knowledge Distillation.

Reviewer Service

- 2026: NeurIPS, ECCV, ICML
- 2025: ICRA, IROS
- 2024: ICRA, IROS

Peer-reviewed Publications

- Minsu Kim, Dongyeun Lee, and Junmo Kim, “Repurposing Influential Observations for Quantization of 3D Object Detection Models,” (*Under Review*), 2026.
- Minsu Kim, Dongyeun Lee, Jaemyung Yu, Jiwan Hur, Giseop Kim, and Junmo Kim, “Inlier-Centric Post-Training Quantization for Object Detection Models,” *International Conference on Learning Representations (ICLR)*, 2026.
- Minsu Kim, Giseop Kim, and Sunwook Choi, “Addressing Diverging Training Costs using BEVRestore for High-resolution Bird’s Eye View Map Construction,” *IEEE Robotics and Automation Letters (RA-L)* with *ICRA*, 2024.
- Minsu Kim, Giseop Kim, Kyong Hwan Jin, and Sunwook Choi, “BroadBEV: Collaborative LiDAR-camera Fusion for Broad-sighted Bird’s Eye View Map Construction,” *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.
- Minsu Kim, Yongjun Lee, Woo Kyoung Han, and Kyong Hwan Jin, “Learning Residual Elastic Warps for Image Stitching under Dirichlet Boundary Condition,” *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024.
- Minsu Kim, Jaewon Lee, Byeonghun Lee, Sunghoon Im and Kyong Hwan Jin, “Implicit Neural Image Stitching With Enhanced and Blended Feature Reconstruction,” *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024.
- Minsu Kim, Taewoong Kang, Dongwoon Song, Seung-Joon Yi, “Development of a Small-Sized Intelligent Home Service Robot,” *International Conference on Ubiquitous Robots (UR)*, 2021.

Awards

- **Gold Prize (Rank 1st)**, National Robot Contest STEAMCUP Home Service Challenge, 2020.
 - Topic: Real-time control of a mobile manipulation robot.
 - Keyword: Grid-based SLAM, Navigation, 4-DoF Manipulation, AprilTag-guided perception.
 - Remarks: Sole Developer of Robot S/W in a team.
- **Grand Prize (Rank 2nd)**, PNU LINC+ Contest of Capstone Design, 2020.
 - Topic: Real-time control of a mobile manipulation robot with an intelligent robotic system.
 - Keyword: Grid-based SLAM, Navigation, 4-DoF Manipulation, 2D Detection, 3D geometry.
 - Remarks: Sole Developer of Robot S/W in a team.
- **Gold Reviewer (Top 25%)**, International Conference on Machine Learning (ICML), 2026.
 - Topic: Good reviewer service evaluated by area chairs.
 - Keyword: 3D perception and model compression.